CV Optimisation Plan

Target: Big Data & Analytics working-student positions (Werkstudent), Munich

Three modifications to the current CV

1. Add a "Projects" section with this research project as the lead entry, written with numbers. Recruiters for analytics roles look for evidence of end-to-end work. Put the project directly under Education and use the pattern action verb + tool + scale + result:

> NLP pipeline for 515,000 hotel reviews (Python, pandas, scikit-learn, gensim): cleaned and labelled 863k text segments; Logistic Regression reached macro-F1 0.93 (vs. 0.92 Naive Bayes, 0.93 Linear SVM).
> Fitted sentiment-specific LDA topic models (13 and 9 topics, selected by coherence) and quantified which themes drive complaints (rooms 30%, breakfast 19%) versus praise (location 30%).
> Wrote a 20-page IEEE-style paper and published reproducible code on GitHub (link).

Keep the bullets to 2-3 lines each, put the strongest number first, and remove coursework entries that do not involve data, code or analysis.

2. Replace the generic skills list with a keyword-matched "Technical skills" block.

Group skills the way job adverts do, and copy the exact terms from each advert (many employers screen CVs with keyword filters). Suggested structure:

> Programming & data: Python (pandas, NumPy, scikit-learn), SQL, Git/GitHub
> Analytics & ML: NLP (TF-IDF, LDA), classification, evaluation (F1, coherence), EDA
> Visualisation & BI: Matplotlib/Seaborn, Power BI gonly if you can demo it), Streamlit
> Big data (learning): Spark/PySpark fundamentals, working with 500k- row datasets
> Languages: English (level), German (level, CEFR), other

List only what you can discuss in an interview. Mark tools you are still learning as "learning" instead of leaving them out or overstating them.

3. Fix the header and add availability, links and a German-market layout.

Add LinkedIn and the GitHub profile (pin this repository), your enrolment status and expected graduation, and a one-line availability statement, e.g. "Available 15-20 h/week during the semester, full-time in semester breaks, from /month/". Working students must stay enrolled and are normally limited to about 20 hours per week during term, so stating this early removes doubt. Use a one-page reverse-chronological layout, a consistent date format and, if the advert is in German, a German-language CV (Lebenslauf). Drop personal details that are not required, and export as PDF named `Lastnane_Firstnane_CV. pdf`.

Application Strategy (1 page)

1. Target roles

Search terms (English and German): Working Student Data Analyst, Werkstudent Data Science / Iachine Learning, Werkstudent Business Intelligence, Werkstudent Datenanalyse gm/w/d), Werkstudent Data Engineering, plus NLP, Customer Analytics, Iarketing Analytics.
Best fit for this profile: analytics / data science with text or customer data; second choice: BI and reporting; stretch: data engineering.

2. Target employers in the Munich area

Large employers with data teams: BMW Group, Siemens, Allianz, Munich Re, Infineon. Product and tech companies: Celonis, Check24, Flix, Sixt. Add a few smaller companies and start-ups from LinkedIn, StepStone and Indeed alerts, and the careers pages and job boards of your university. Consulting and travel or hospitality-tech firms suit the customer-feedback focus of the project.

3. Core skills to highlight

Python and SQL for analysis, shown by the repository and notebooks (cleaning, modelling, evaluation).
Working with large, messy data: 515k records, ETL, memory- and time-aware design (sampling, saved checkpoints).
Applied ML and NLP: classifier comparison with macro-F1, topic modelling with coherence-based model selection.
Communicating results: charts, a written IEEE-style paper, and (optionally) a Power BI or Streamlit dashboard.
Good practice: Git, documented README, fixed random seeds, awareness of data protection (GDPR).

4. How the research paper strengthens the profile

It is concrete evidence instead of a claim: a public repository and a PDF that a recruiter can open in two minutes. It shows the full analytics cycle (question, data, method, result, business recommendation), it is domain-relevant for customer-feedback and service analytics, and it provides interview material, for instance why labels derived from the review form are a limitation, or why separate topic models were used.

5. Action plan

Days 1-2: finish CV (three changes above), tidy LinkedIn, pin the repository, add the paper link.
Week 1: set up alerts; apply to 8-10 roles with a short, tailored cover letter (3 paragraphs: motivation for the company, the project as evidence, availability).
Weeks 2-4: 5-8 applications per week; send a polite follow-up after 7-10 days; ask lecturers or contacts for referrals.
Interview preparation: be able to explain TF-IDF, LDA, macro-F1 and your limitations in plain language; practise basic SQL joins and aggregations and one Python data-cleaning exercise.

Success measures: applications sent per week, interview invitations, and feedback collected after each interview.